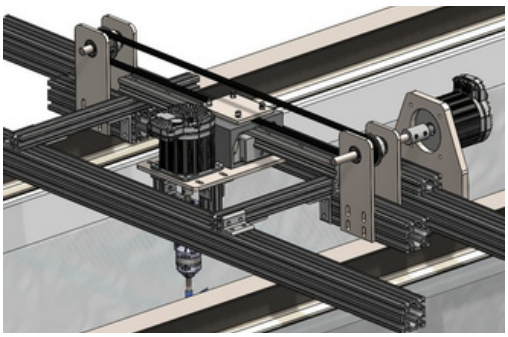
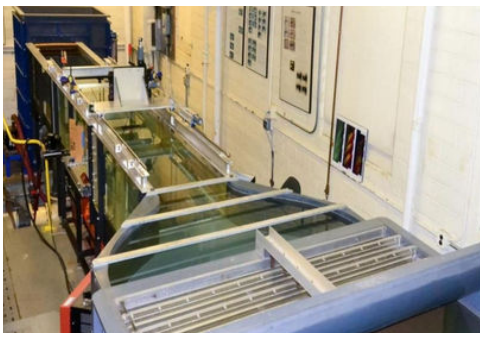
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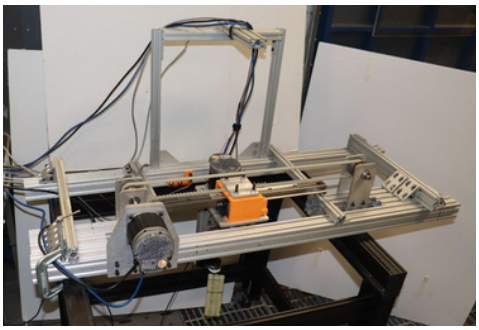
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CYBER-PHYSICAL ROBOTIC FLAPPER FOR STUDYING BIO-PROPULSION







What?

- Designed and built a robotic swimmer for a water channel to study thrust and energy generation from ocean waves.
- Made pitch and heave cyber-physical: the rig physically replicates programmed mass, stiffness, and damping
- Foil parameters can change in seconds instead of swapping springs or reassembling hardware, saving setup time and cost.

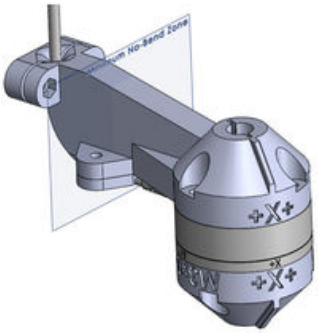
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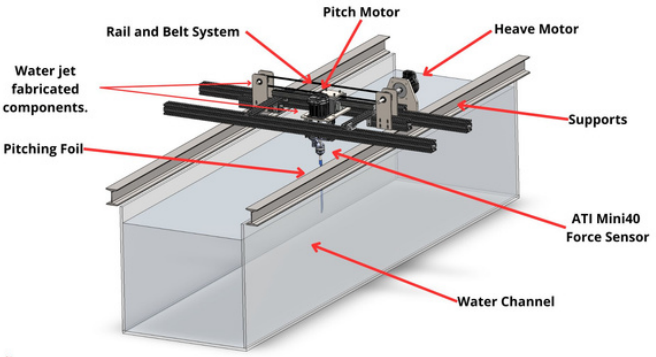
- Designed in SOLIDWORKS; built with CNC, waterjet, and 3D printing. GD&T on all drawings.
- Force feedback closes the loop: a 6-axis sensor measures fluid force on the foil, the controller solves the mass-spring-damper equation with the programmed values at 4 kHz, and the servo moves the foil to that response, so it behaves as if a real spring and damper were attached.
- Built a Simulink control loop to acquire data, automate experiments, and ensure safe operation.

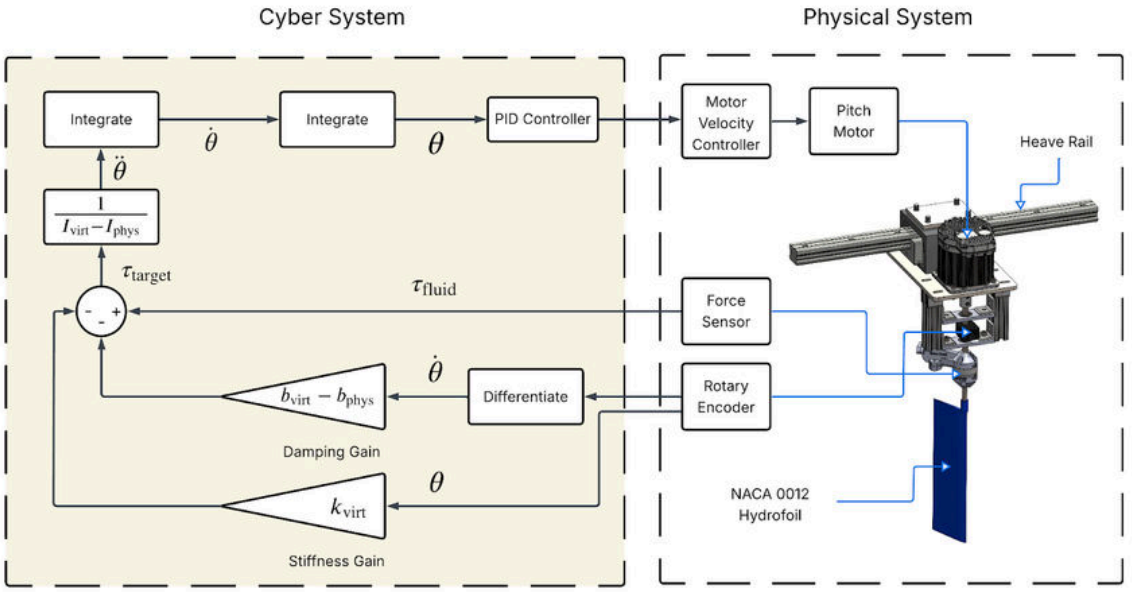
Results

- Automated the full test cycle: sweeps parameters and saves every data set, cutting test time.
- Diagnosed motion error caused by mechanical backlash and redesigned linear-drive components with a pulley-belt system, reducing positioning error 5×.
- Control tuning cut operational noise by 80%.
- Built-in safety limits protect hardware and operators.







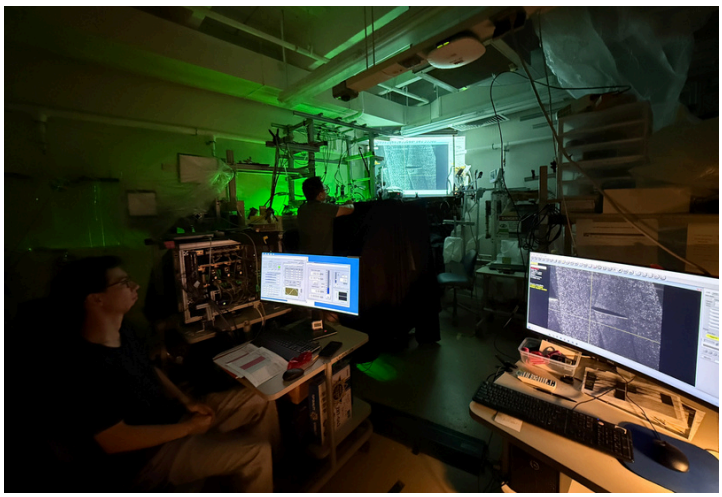


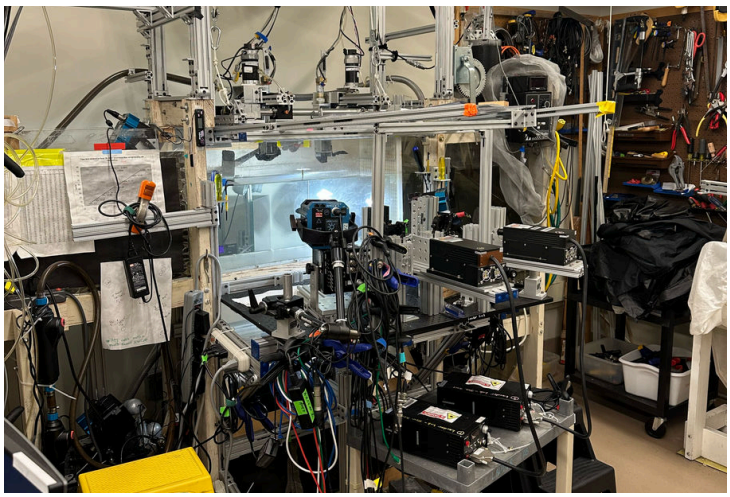
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FISH ROBOT SWIMMING EFFICIENCY - HARVARD UNIVERSITY





What?

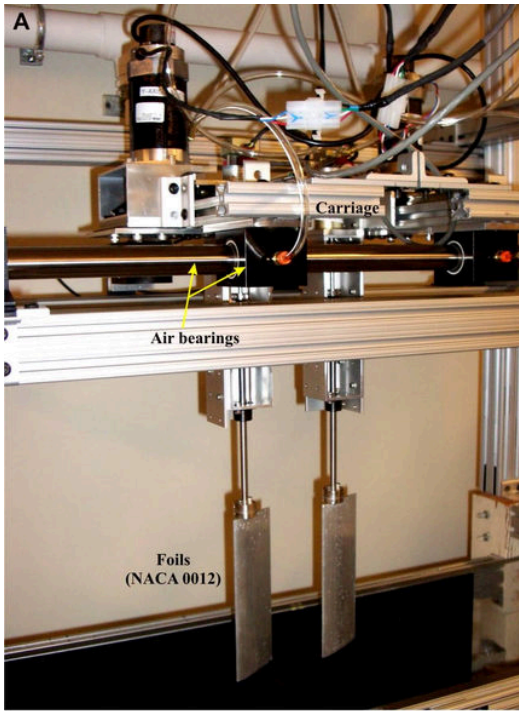
- Studied how a fish swimming in line behind a leading swimmer gains thrust from its wake without actively swimming, with applications to ships and underwater drones.
- Measured how frequency, pitch amplitude, and flow speed affect swimming and station-holding performance.

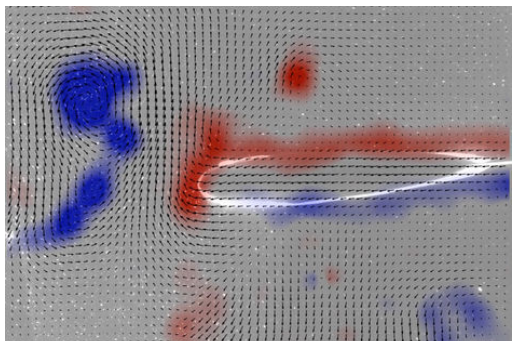
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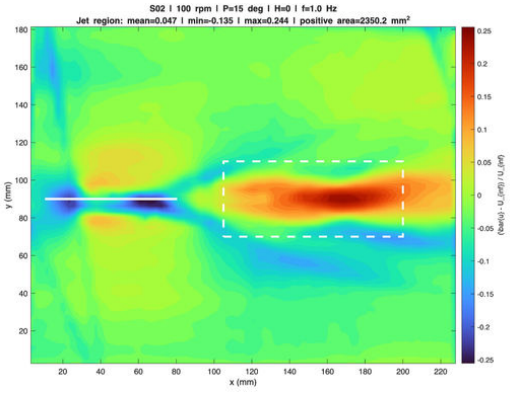
- Operated a robotic flapper with air bearings, LabVIEW, PID control, and NI-DAQ.
- Ran hydrofoil and passive-fish experiments across flow conditions.
- Performed particle image velocimetry (PIV); processed wake data in MATLAB and PIVlab.
- Built 2-D tracking and 3-D reconstruction pipelines in MATLAB and R.

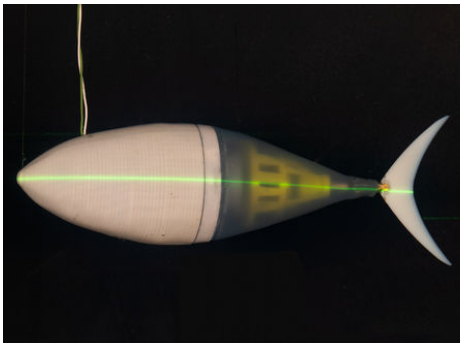
Results

- Presented findings at the ONR MURI Annual Review Meeting (Sep 2026) and the APS Division of Fluid Dynamics Meeting (Nov 2026).
- Quantified swimming kinematics across flow conditions, which can inform ship propulsion and underwater drone design.











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CARBON FIBER SUSPENSION SYSTEM FOR FORMULA SAE





What?

- Designed and manufactured a carbon fiber–aluminum hybrid suspension control arm for a Formula SAE vehicle.
- Replaced steel members to reduce mass while maintaining required load capacity.

How?

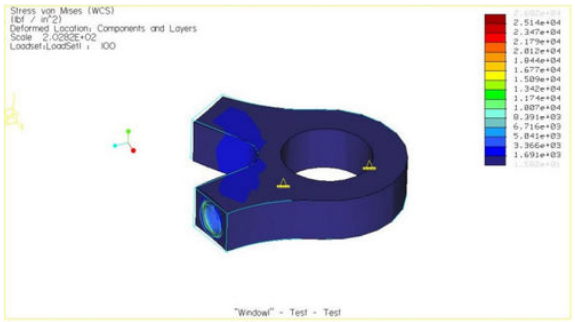
- Modeled geometry in SolidWorks and conducted FEA to evaluate stress and deflection under braking and cornering loads.
- Machined aluminum end joints and bonded them to carbon fiber tubes using structural epoxy.

Results

- Reduced component mass compared to steel baseline.
- Measured strain and load response were consistent with FEA predictions.
- Component sustained expected loads without material failure or joint separation.









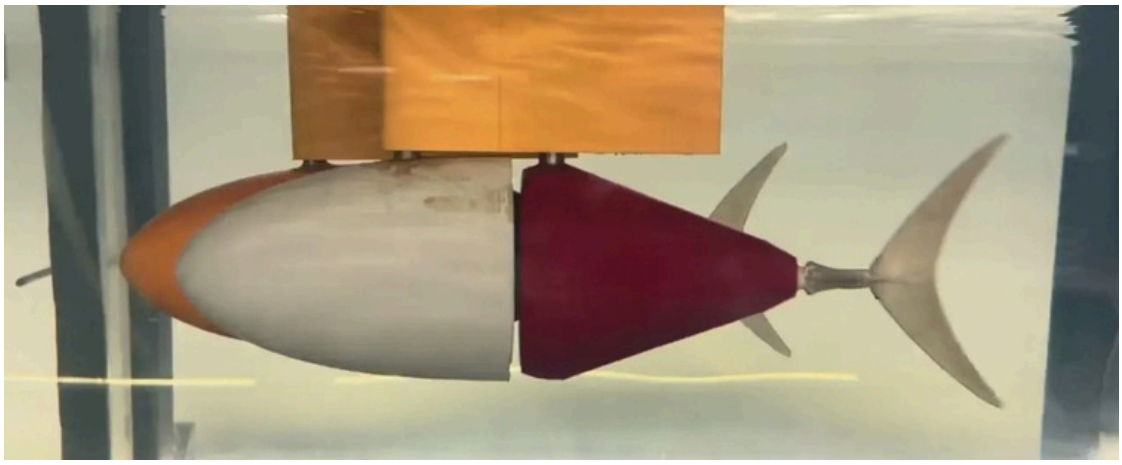


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TUNABOT SWIMMING EXPERIMENTS



What?

- Two TunaBot robotic fish swim in line in a water channel (leader and follower) to study how the follower interacts with the leader's wake.
- Focus on the Knoller–Betz effect: how a flapping wake produces thrust and how a follower can harvest it.

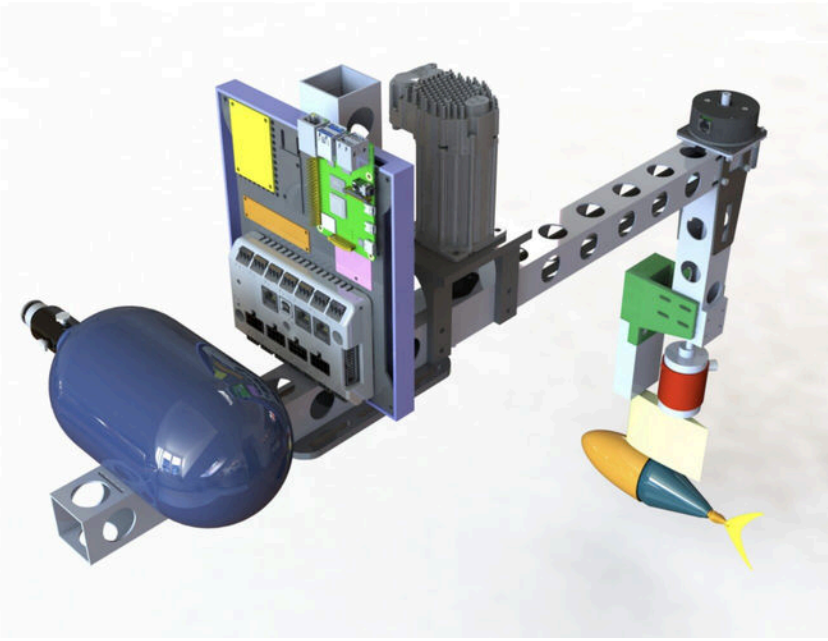
How?

- Operated both robots through Arduino, with MATLAB/Simulink and an NI DAQ for synchronized control and data acquisition.
- Ran in-line trials with in-phase and anti-phase tail beats.

Results

- Used PIV to capture the wake between the leading and trailing swimmers; processed flow fields in MATLAB/PIVlab.
- Compared data to 3D printed passive fish experiments.



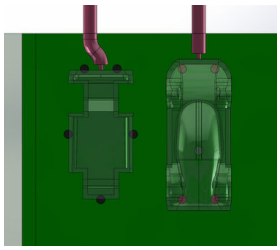


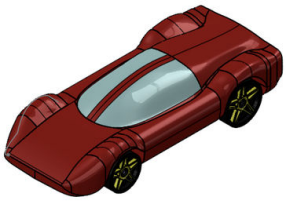
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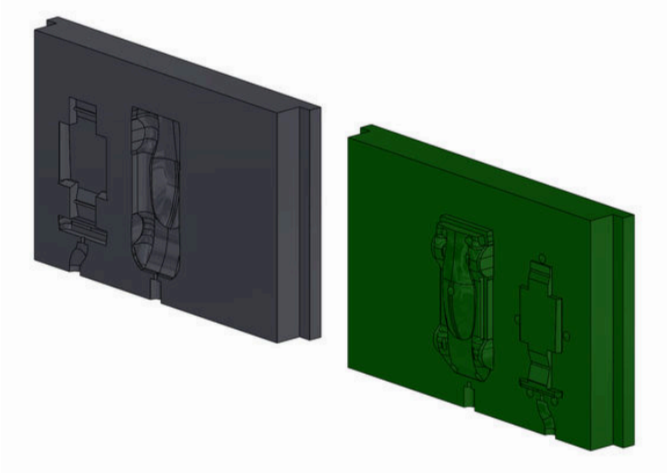
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RACE CAR INJECTION MOLDING PROJECT









What?

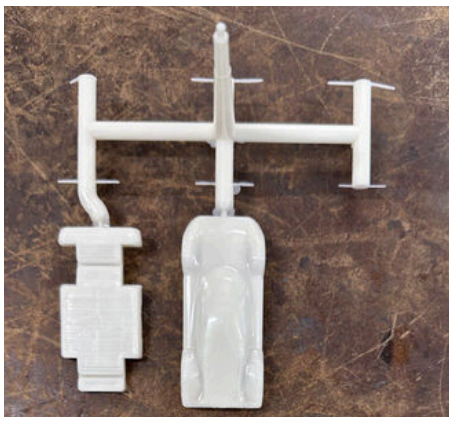
- Designed and manufactured a two-part aluminum injection mold for a race car body component.
- Developed a complete molding workflow including part design, gating, and runner system.

How?

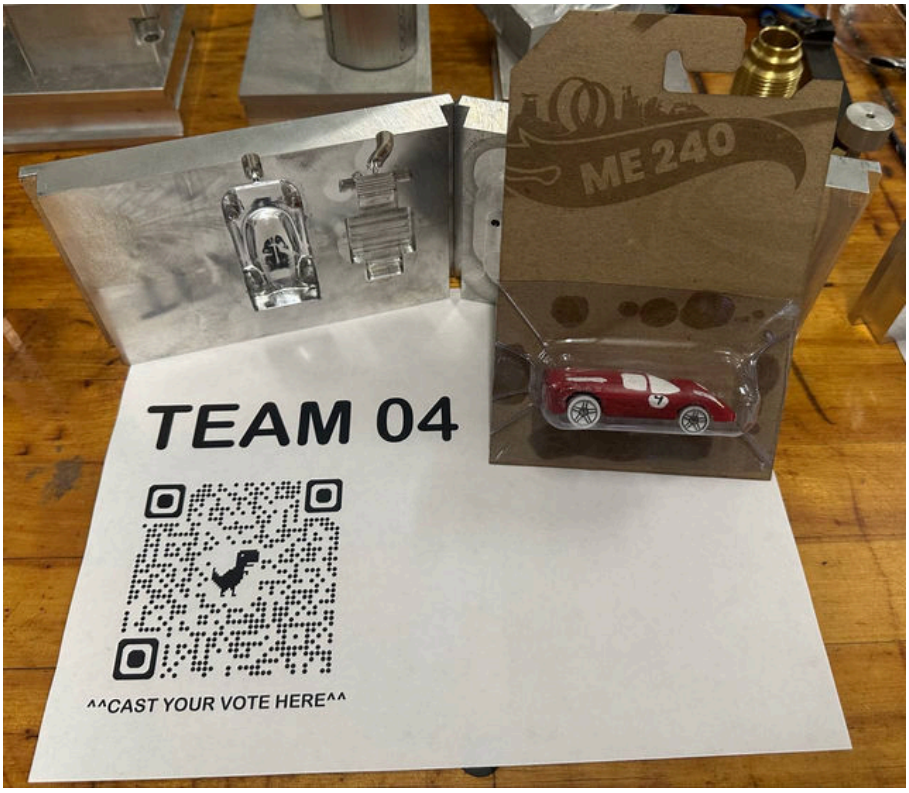
- Modeled part and mold assembly in SolidWorks.
- Fabricated mold using CNC machining and waterjet cutting.
- Applied GD&T to ensure dimensional accuracy and proper part ejection.
- Conducted molding trials and evaluated part quality and repeatability.

Results

- Successfully produced molded race car components with consistent geometry and surface finish.
- Achieved proper mold alignment and clean part release.
- Validated manufacturability through multiple successful production cycles.





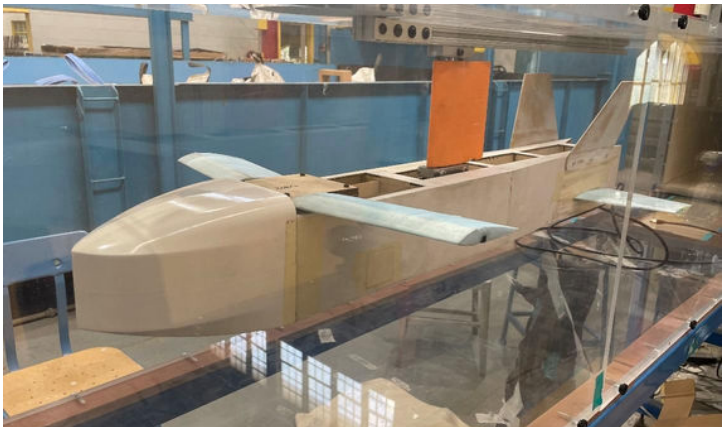


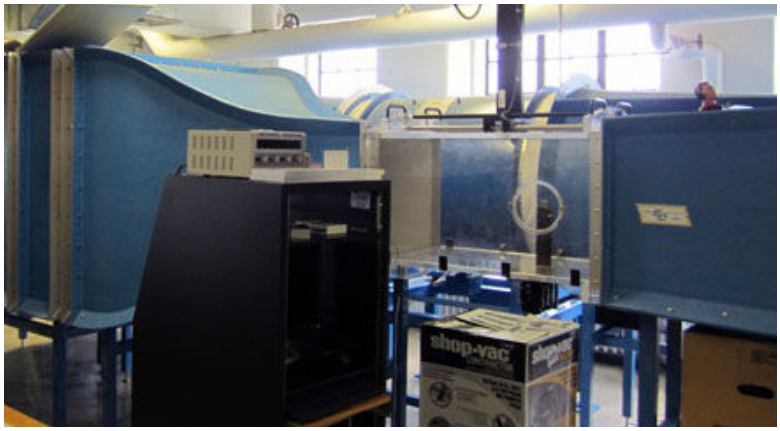
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WIND TUNNEL DRONE TESTING - AEROTARGETS INTERNATIONAL





What?

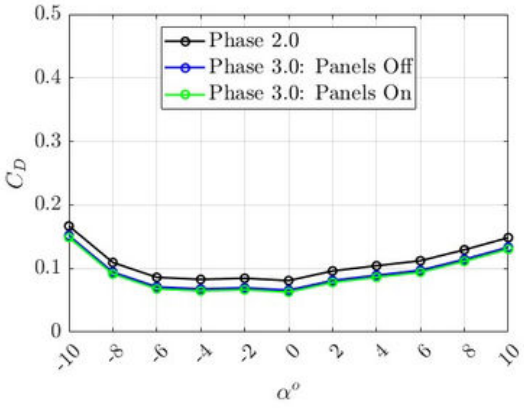
- Tested a scaled-down military drone in a wind tunnel to measure its aerodynamic performance.
- Split the experiment into 3 phases focusing on (1) preliminary aerodynamic stability studies, (2) a scaled up model at a higher Re used for extensive drag tests with panels off and (3) with panels on.

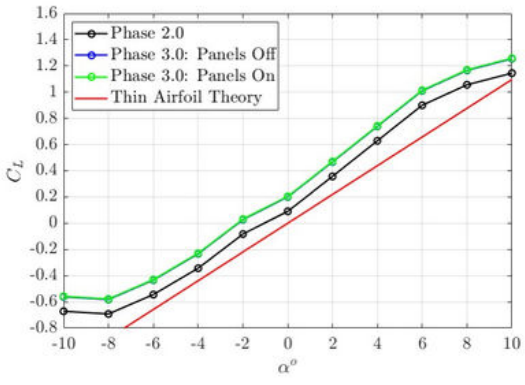
How?

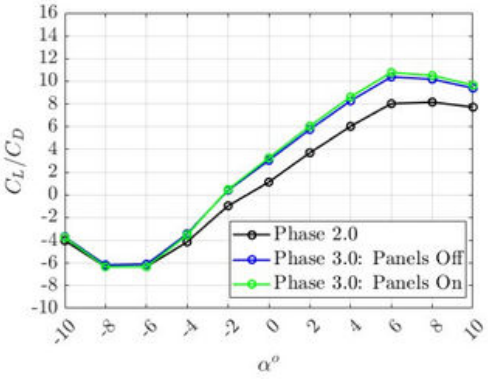
- Built a four-bar linkage structure that allows every test angle to be set without re-rigging.
- Used SolidWorks to scale down and alter model for compatibility with force sensor / wind tunnel.
- Utilized an NI DAQ card and MATLAB to acquire and process data.

Results

- Phase 1 revealed that the model is dynamically stable in both pitch and roll motions around its CG and weakly stable in yaw.
- Phase 2 and 3 showed a lower drag coefficient, increasing its L/D ratio.
- Communicated with AeroTargets engineers to recommend fuselage design modifications, achieving a 10% drag reduction.







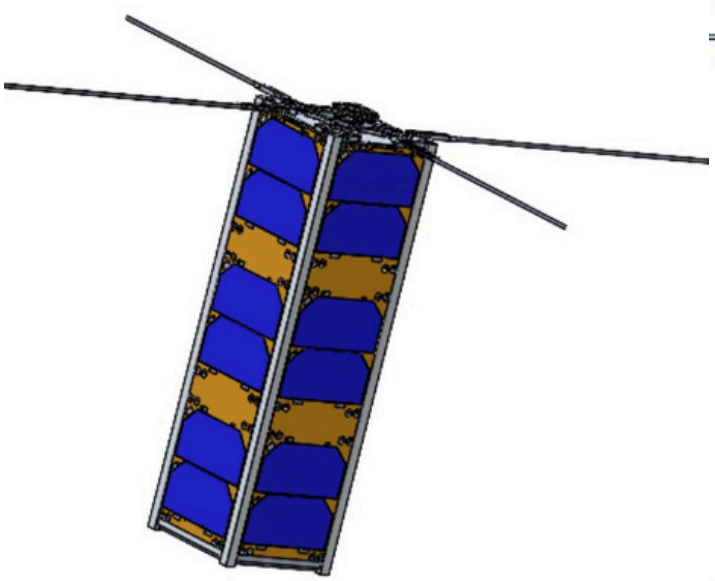
AEROTARGETS INTERNATIONAL

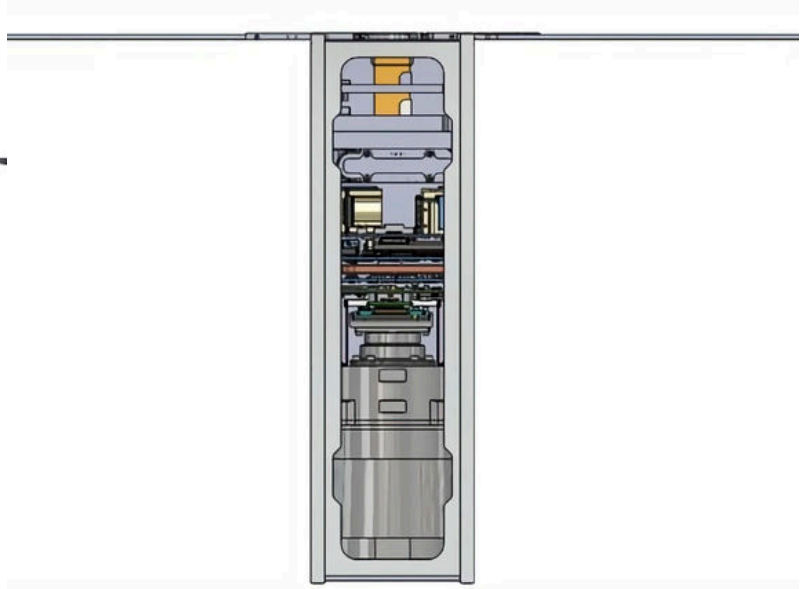
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OSPREY CUBESAT - LEHIGH UNIVERSITY'S FIRST NANOSATELLITE





What?

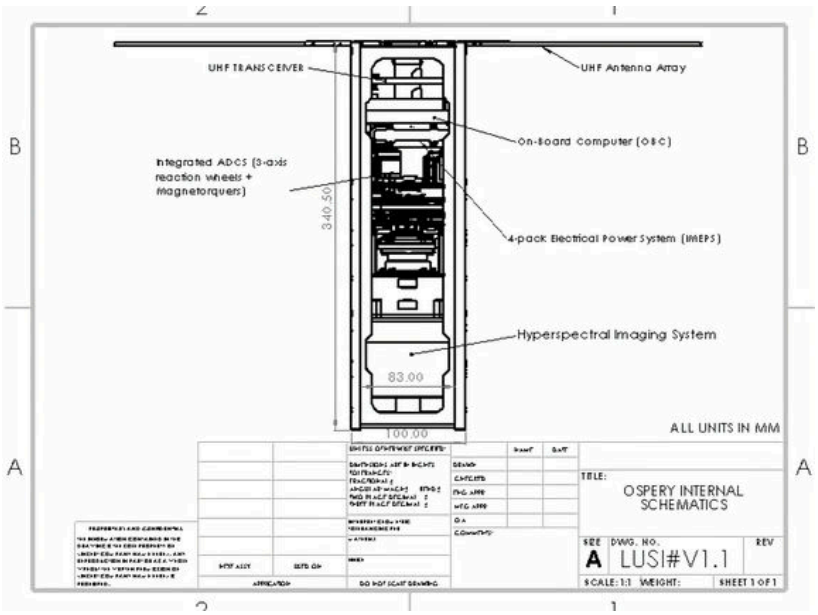
- Led the design of OSPREY, Lehigh University's first CubeSat, a 3U satellite built to spot ocean plastic from orbit.
- Specified a VS/SWIR camera system able to resolve plastic within a 1-square-meter area.

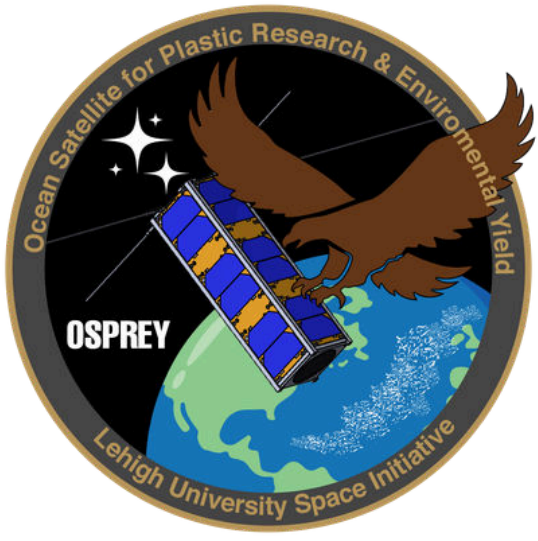
How?

- Led CAD modeling in SolidWorks, optimizing the center of mass and moments of inertia.
- Used GD&T principles, applied material selection, mass budgeting, to ensure tolerance and compliance with NASA standards.
- Managed a team of 6 students and coordinated with other sub-teams to integrate deployable solar cells and antenna systems

Results

- Raised $150,000 in funding from Lehigh alumni and sponsors.
- Optimized the mass budget by 23% and increased payload volume by 33%.
- Successfully submitted a proposal for NASA's 2024 CubeSat Student Launch Initiative.



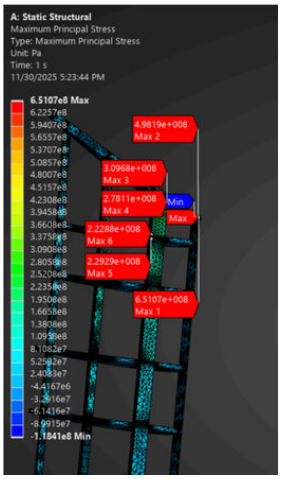


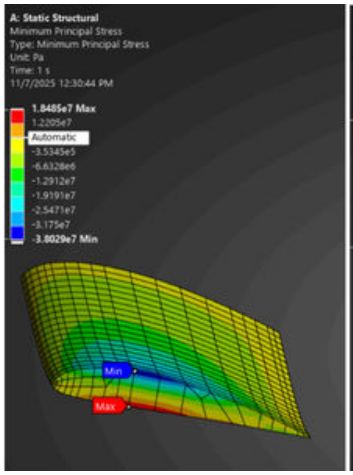
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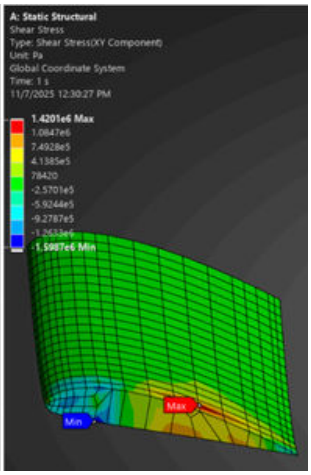
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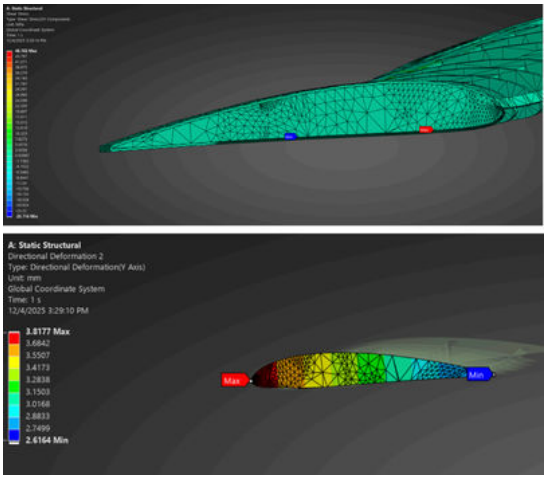
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AEROELASTIC WING DESIGN - LEHIGH UNIVERSITY









What?

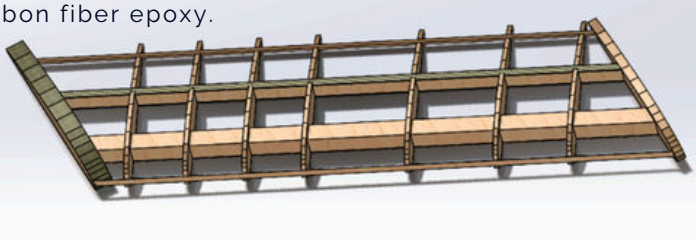
- Designed and tested three balsa wood wings to find how much weight a wing can lose before it becomes unstable in flight.
- Compared straight, swept-forward, and rib-and-strut designs (Gottingen 398 airfoil).
- Evaluated torsional stiffness, aeroelastic coupling, and divergence speed under a 100 N/m distributed lift load.
- Fabricated rib–strut structure using a laser cutter and reinforced critical regions with carbon fiber epoxy.

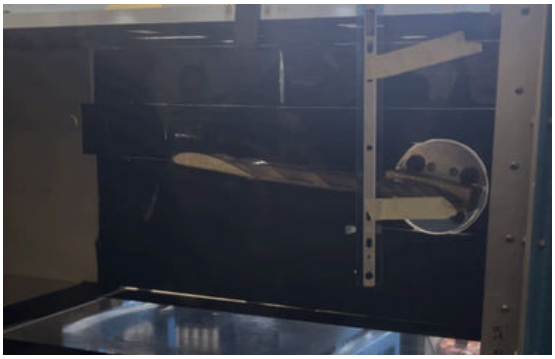
How?

- Modeled geometry in SolidWorks.
- Performed structural analysis using ANSYS Workbench FEA to extract deflections and stresses.
- Computed twist, torsional stiffness, shear center, and divergence speed in MATLAB.
- Conducted wind tunnel testing.

Results

- Showed that aggressive weight reduction (to 0.247 N) led to a ~98% stiffness drop, lowering divergence speed to 26.2 m/s.
- Identified the base shear interface as the critical failure location during wind tunnel testing (~90 mph).
- Established quantitative relationship between mass reduction, stiffness penalty, and aeroelastic stability.





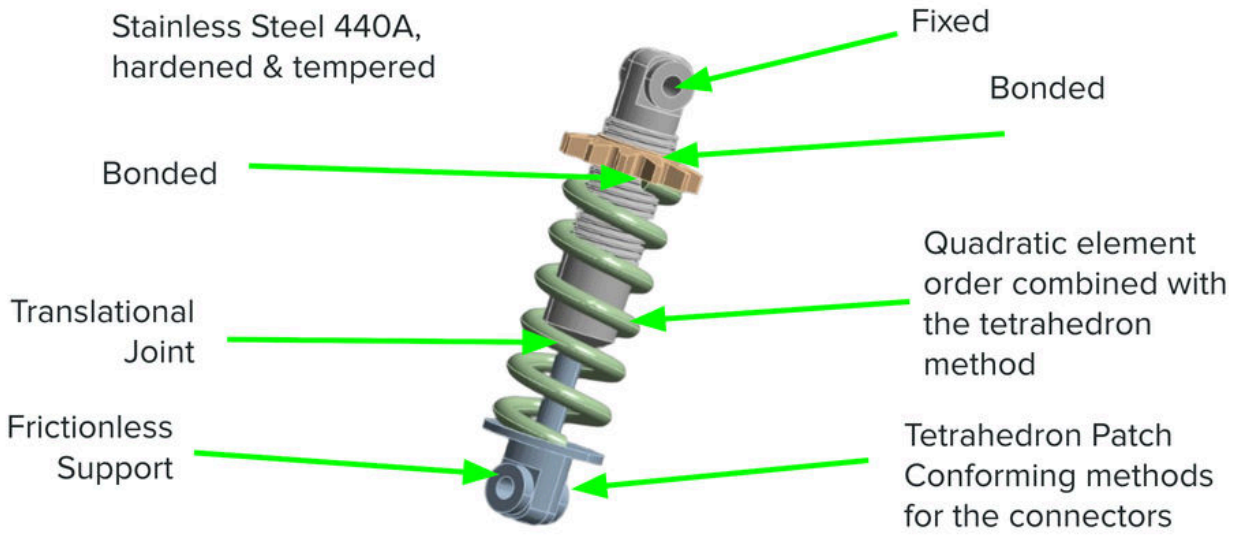


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SHOCK ABSORBER - FEA MODAL ANALYSIS



What?

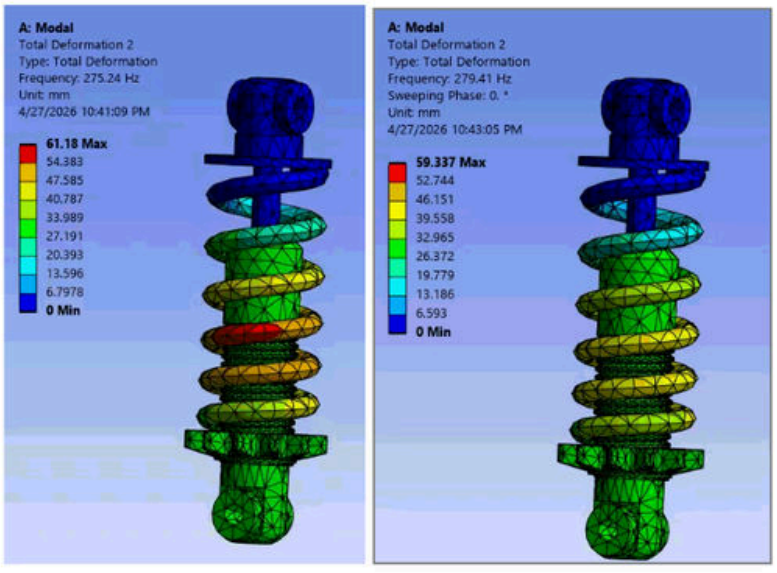
- Performed modal analysis on a scaled shock absorber assembly to study its natural vibration behavior.
- Tested how damping affects vibration decay and stability.

How?

- Set up the model in ANSYS with fixed, bonded, frictionless, and translational joint boundary conditions.
- Used Stainless Steel 440A and quadratic tetrahedral mesh elements.
- Tested damping ratios from 0.05 to 0.3.

Results

- Natural frequencies stayed around 152–712 Hz.
- Higher damping caused faster vibration decay without major frequency shifts.
- Results showed the system is mainly mass–stiffness dominated.
- Fixed contact issues by improving boundary conditions and using a translational joint where sliding motion was needed.



0.05 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1.	151.89	-1.551	1.0211e-002	-6.4161e-002
2.	355.23	-5.3157	1.4963e-002	-9.4023e-002
3.	367.36	-5.5089	1.4994e-002	-9.4222e-002
4.	414.7	-6.1958	1.4939e-002	-9.3874e-002
5.	419.73	-6.2565	1.4904e-002	-9.3658e-002
6.	704.77	-10.564	1.4988e-002	-9.4183e-002

0.1 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1.	152.02	-5.1625	3.3939e-002	-0.21337
2.	355.63	-17.699	4.9706e-002	-0.3127
3.	367.77	-18.342	4.9811e-002	-0.31336
4.	415.17	-20.629	4.9627e-002	-0.3122
5.	420.2	-20.832	4.9515e-002	-0.31149
6.	705.59	-35.176	4.9791e-002	-0.31323

0.2 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1.	152.46	-10.276	6.7251e-002	-0.42351
2.	356.94	-35.268	9.8328e-002	-0.62082
3.	369.13	-36.549	9.8533e-002	-0.62213
4.	416.7	-41.106	9.817e-002	-0.61981
5.	421.73	-41.512	9.7959e-002	-0.61847
6.	706.23	-70.097	9.8492e-002	-0.62187

0.3 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1.	153.16	-15.297	9.938e-002	-0.62753
2.	359.06	-52.587	0.14491	-0.92021
3.	371.34	-54.498	0.14521	-0.92214
4.	419.19	-61.291	0.14467	-0.91867
5.	424.22	-61.901	0.14439	-0.91682
6.	712.5	-104.52	0.14515	-0.92175

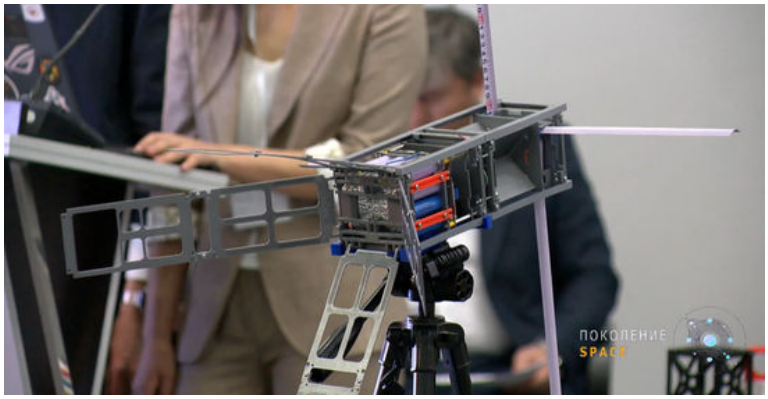
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SKILLSAT STARTUP





What?

- Founded SkillSat, an EdTech startup to manufacture STEM kits that teach students to build stratospheric satellites.

How?

- Used Arduino and Raspberry Pi as the base for satellite systems.
- Utilized additive manufacturing to prototype 25+ designs.
- Applied DFA and DFM principles to reduce production costs by 44%.
- Tested prototypes at the National Space Center of Kazakhstan.

Results

- Launched satellites to Near Space (40 km+) via stratospheric balloons.
- Raised $15K in funding and distributed 80+ kits to 240+ students across 25+ schools.
- Recognized as a Top-3 SpaceTech and Top-15 EdTech Startup by Forbes Kazakhstan.



